

Guided Lab 4.1 — Instructions

Course 02 – Introduction to Machine Learning for Cybersecurity

Module 04: Model Evaluation and Tuning

Lab Goal

In this guided lab, students will evaluate the performance of supervised machine learning models using common classification metrics and model evaluation techniques.

Students will:

- Train a machine learning classification model
- Generate predictions
- Interpret confusion matrices
- Calculate evaluation metrics
- Explore cross-validation techniques
- Understand how model evaluation supports cybersecurity and fraud detection systems

Important LabNote

A model with high accuracy may still perform poorly on fraud detection tasks if fraudulent transactions are rare.

This lab focuses on understanding:

- Precision
- Recall
- False positives
- False negatives
- Model reliability

in cybersecurity and fraud detection systems.

Dataset

For this lab, we will use the:

✓ **Credit Card Fraud Detection Dataset** ([creditcard.csv](#))

[creditcard.csv](#)

Dataset Information

Column	Description
V1–V28	Anonymized transaction features
Time	Transaction time
Amount	Transaction amount
Class	Target label (0 = Normal, 1 = Fraud)

✂ Guided Lab Activities

Part 1 — Load and Explore the Dataset

Students will:

- Load the dataset
- Review class distribution
- Explore dataset structure

Part 2 — Prepare the Data

Students will:

- Define features and labels
- Split training and testing data
- Scale features if needed

Part 3 — Train a Classification Model

Students will:

- Train a supervised learning model
- Generate predictions

Example Models:

- Logistic Regression
- Decision Tree
- Random Forest

Part 4 — Evaluate Model Performance

Students will:

- Calculate accuracy
- Generate confusion matrices
- Calculate precision, recall, and F1 score
- Interpret model performance

Part 5 — ROC Curves and AUC-ROC

Students will:

- Generate ROC curves
- Interpret AUC-ROC values
- Compare model effectiveness

Part 6 — Cross-Validation

Students will:

- Apply K-Fold Cross-Validation
- Compare cross-validation results
- Understand model generalization

Part 7 — Introductory Hyperparameter Tuning

Students will:

- Adjust basic model settings
- Compare performance changes
- Observe how tuning affects results



Submission Requirements

Submit:

- ✓ Completed Jupyter Notebook (.ipynb)
- ✓ All code cells executed with outputs visible
- ✓ Evaluation metrics and visualizations included
- ✓ Written interpretation of results